Meridian Health Group — Patient Experience Team: Domain Context

Team Scope

The Patient Experience Team owns the **patient-facing digital portal** — a Progressive Web App (PWA) with native iOS/Android wrappers — covering appointment booking, medical records access, repeat prescription requests, secure messaging with care teams, proxy access for carers, and co-payment collection. All clinical data is read from and written back to Epic EHR via the FHIR Adapter.

Team Personas

Persona	Description	Key Needs
Patient	Registered individual receiving care at Meridian	Book and manage appointments, view lab results, request prescriptions, message their care team
Carer / Guardian	Family member or guardian managing a dependent's care; requires proxy access	Book appointments, view records, send messages on behalf of the patient; same access as patient but clearly labeled as proxy

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Appointment Management

Appointment Types Available to Patients

Type	Lead Time	Notes
GP Appointment	Bookable up to 4 weeks ahead	Standard 10–15 min slot; bookable directly by patient
Telehealth	Same as GP	Video consultation via Doxy.me; patient joins from portal
Diagnostic / Imaging	After referral	Radiology or pathology; booked after GP referral; slot selected by patient from available windows
Urgent / Walk-in	Same-day only	Queue-based; no advance booking; patient can see estimated wait time

- **Specialist Referral** bookings: initiated by admin staff on behalf of the patient after a GP referral; patient is notified when a slot is confirmed (not self-service in Phase 1)

Booking Rules

- Cancellation within **2 hours** of appointment triggers a no-show flag in Epic (written back via FHIR `Appointment.status = noshow`)
- Rescheduling counts as cancellation + rebook; same 2-hour rule applies
- Automated reminders: SMS + email 24 hours before and 2 hours before each appointment
- Waitlist: if all slots for a clinic are booked, patient can join a waitlist; notified if a slot opens up

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Medical Records Access

- Patients can view records for the **last 5 years** (older records available on request via branch/admin)
- **Lab results**: displayed within 4 hours of Epic verification (polling every 15 minutes via FHIR Adapter)
- **Radiology reports**: delivered as PDFs via the same FHIR pipeline (polling every 30 minutes)
- **Medication list**: current active medications from Epic `MedicationRequest` resources
- **Care plans**: active care plans from Epic `CarePlan` resources; read-only view for patient
- **Record download**: patient can download a PDF summary of their health record (HIPAA Right of Access)
- **Corrections**: patient can request corrections to demographic and contact information; request goes to admin staff for review (not auto-applied)

Repeat Prescription Requests

- Patient submits a repeat prescription request via the portal (minimum 7 days before medication runs out)
- Request lifecycle: Submitted → GP Reviewed → Approved / Rejected → Dispensed
- Patient sees status in real time (polled from Epic via FHIR Adapter)
- Approved prescriptions: patient is notified when prescription is ready for collection at their nominated pharmacy
- **Controlled substances cannot be requested via the portal** — legislative constraint; patient shown a message directing them to contact their GP directly
- Medication reminders: opt-in; patient sets their own schedule; aligned to prescription frequency

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Secure Messaging

- Patients can send asynchronous text messages and attachments to their care team (GP or specialist)
- Attachments: photos < 5 MB, PDFs < 10 MB; stored in S3 SSE-KMS; delivered via pre-signed URL (5-minute TTL)
- Response SLA: **2 business days** for non-urgent messages; displayed as a banner on the messaging screen
- **Urgent escalation**: Messaging Service detects an urgent flag (set by clinical staff on reply) and triggers an immediate Twilio SMS to the patient
- Message history: full thread visible to patient; stored as FHIR `Communication` resources in Epic
- **Messages become part of the clinical record** — patients are informed of this before sending

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Proxy / Carer Access

- A patient may grant a carer or guardian proxy access to their HealthConnect account
- **Consent flow**: patient uploads a scanned consent form; Admin staff manually approves the proxy grant (not auto-approved)
- Proxy can: book appointments, view records (up to 5 years), view and send messages on behalf of the patient
- All proxy actions are logged in the Audit Service with both `patientId` and `proxyUserId`
- A patient may revoke proxy access at any time from their account settings; revocation takes effect immediately

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Co-Payment Collection

- Certain outpatient visit types require a co-payment collected at the time of booking (via Stripe)
- Payment is required to complete the booking; failed payment blocks booking completion and shows a retry prompt
- Payment confirmation email sent via SendGrid; digital receipt stored against the appointment record
- **Inpatient billing is handled entirely by Epic** — out of scope for HealthConnect

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Account & Identity

- Registration: NHS/MRN number + date of birth for identity verification
- MFA: mandatory — SMS OTP or TOTP authenticator app
- Account lockout: 5 consecutive failed logins; self-service unlock via email verification
- Session timeout: 15 minutes (web), 30 minutes (mobile)
- **Consent management**: granular opt-ins for clinical communications, research participation, and marketing — stored in Cognito user attributes and respected by the Notification Service

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Business Rules Specific to Patient Experience

1. Cancellations within **2 hours** trigger an Epic no-show flag (FHIR write-back); patients who accumulate 3 no-shows in 12 months are flagged to admin staff
2. Lab results are displayed to the patient **only after Epic verification** — unverified or interim results must not be shown
3. Controlled substance prescription requests: the portal must present a clear message ("This medication cannot be requested online — please contact your GP directly") before the patient types any request
4. Proxy access must require **admin approval** before the proxy can access the patient's account — no self-service grant
5. Co-payment must be collected **before the booking is confirmed** in Epic; if the booking confirmation to Epic fails, the Stripe payment is automatically refunded
6. All message attachments must be **virus-scanned** (ClamAV lambda on upload) before being stored or delivered